

Hyunsoo Kim

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EDUCATION

The University of Texas at Austin

Austin, TX

Ph.D. in Electrical and Computer Engineering

Sep 2026 – Present

- Co-advised by Prof. Guandao Yang and Prof. Diana Marculescu
- Research: Leveraging image and video generation models for visual understanding and perception

Korea University

Seoul, Republic of Korea

M.S. in Artificial Intelligence

Sep 2024 – Aug 2026

- Co-advised by Prof. Donghyun Kim and Prof. Suhyun Kim
- Research: Diffusion models, data-free synthesis, AI safety, and generative model alignment

Inha University

Incheon, Republic of Korea

B.S. in Statistics and Software Engineering

Mar 2018 – Aug 2024

- Two-year break for military service (Jul 2019 – Feb 2021)

RESEARCH INTERESTS

Image and video generation (diffusion models, flow matching, world models)

Generate to Understand (leveraging generative models for perception, reasoning, and visual understanding)

AI Safety (safe generation, concept erasure, trustworthy AI)

PUBLICATIONS

- [C5] **Consistency-Preserving Concept Erasure via Unsafe-Safe Pairing (PAIR)** NeurIPS 2026
- Yongwoo Kim*, Sungmin Cha*, **Hyunsoo Kim**, Jaewon Lee, Donghyun Kim (*: equal first authorship)
 - Proposed PAIR, a consistency-preserving concept erasure framework that reframes erasure as semantic realignment using unsafe-safe pairs, with Fisher-weighted DoRA initialization for selective suppression while maintaining generation quality
- [C4] **Correlation-Weighted Multi-Reward Optimization for Compositional Generation** ECCV 2026
- Jungmyung Wi, **Hyunsoo Kim**, Donghyun Kim
 - Proposed correlation-based reward reweighting for multi-concept text-to-image generation; dynamically estimates which concepts conflict across generated samples and assigns them stronger optimization signals via GRPO fine-tuning on SD3.5 and FLUX.1-dev
- [C3] **Self-Improving Diffusion Classifiers with Minority Preference Optimization** ECCV 2026
- **Hyunsoo Kim***, Jungmyung Wi*, Soobin Um, Donghyun Kim[†], Suhyun Kim[†] (*: equal contribution, †: equal advising)
 - Introduced MiPO, a self-improving framework that expands diffusion model coverage in low-density regions via GRPO with a novel DDIM-reconstruction-based minority preference reward
 - Improved zero-shot diffusion classification by up to 3.8% across multiple benchmarks without any additional real image data
- [C2] **Diffusion-assisted Data-free Image Synthesis with Domain-Class Alignment** ICML 2025
- Yujin Kim*, **Hyunsoo Kim***, Hyunwoo J. Kim, Suhyun Kim (*: equal contribution)
 - Proposed DDIS, a framework leveraging text-to-image diffusion models for data-free knowledge distillation with explicit alignment of domain and class distributions
- [C1] **Difference Inversion: Interpolate and Isolate the Difference with Token Consistency** CVPR 2025
- **Hyunsoo Kim**, Donghyun Kim[†], Suhyun Kim[†] (†: equal advising)
 - Proposed a token-consistency-based DDIM inversion method that interpolates and isolates visual differences between image pairs, enabling image analogy generation for style transfer, object manipulation, and scene editing

- [P1] **Custom Forcing: Training-Free Subject Customization for AR Video Generation** Preprint
- Yunseung Ok*, **Hyunsoo Kim***, Minseo Kim, Suhyun Kim[†] (*: equal contribution, †: corresponding author)
 - Proposed a training-free method that customizes streaming autoregressive video models by writing reference images into the KV cache, with drift-adaptive value amplification and anchor contrast guidance to keep the subject identity over two-minute rollouts
 - Kept DINO-I between 0.58 and 0.62 over two minutes while fixed anchors fell to 0.42, generating each frame 9.5–28.5× faster than causal image- and reference-to-video baselines

EXPERIENCE

Korea Institute of Science and Technology (KIST) Seoul, Republic of Korea
Graduate Researcher (Advisor: Suhyun Kim) *Sep 2024 – Aug 2026*

- Led research on image analogy generation, safe image translation, and self-improving diffusion classifiers
- Published 2 first-author papers at ECCV 2026 and CVPR 2025; 1 co-first-author paper at ICML 2025
- Developed training pipelines for diffusion model fine-tuning with RLHF/GRPO on multi-GPU clusters

Korea Institute of Science and Technology (KIST) Seoul, Republic of Korea
Undergraduate Research Intern (Advisor: Suhyun Kim) *Mar 2024 – Jun 2024*

- Researched data-free image synthesis by optimizing text embeddings on text-to-image diffusion models
- Contributed to DDIS framework that was later accepted at ICML 2025

Medical Image Innovation Lab, Seoul National University Seoul, Republic of Korea
Undergraduate Intern (Advisor: Wonjin Lee) *Dec 2023 – Feb 2024*

- Researched conditional generation and inpainting in medical imaging using diffusion models
- Implemented and evaluated diffusion-based inpainting pipelines for CT and MRI reconstruction tasks

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, Bash, R
ML Frameworks: PyTorch, Hugging Face (Diffusers, Accelerate, TRL), DeepSpeed, JAX
Tools: Git, Docker, Weights & Biases, SLURM, Linux, L^AT_EX, Jupyter
Research Areas: Diffusion Models, Flow Matching, RLHF/GRPO, Image/Video Generation, Zero-shot Classification, AI Safety, Concept Erasure, Data-free Knowledge Distillation